Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTERB.TECH DEGREE EXAMINATION, JULY2021

Course Code: PHT100

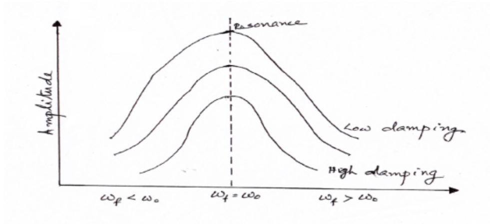
Course Name:ENGINEERING PHYSICS A

PART A

Answer all questions. Each question carries 3 marks

Q1 **What is meant by sharpness of resonance?**

ans **When is increased or decreased from resonance frequency,the amplitude falls from the maximum value.**
The term sharpness of resonance refers to the rate of decrease of amplitude with the change in frequency of the applied periodic forceon either side of the resonant frequency.



Q2 **State the 3 laws of transverse vibrations**

ans **Frequency of transverse vibrations of stretched string**

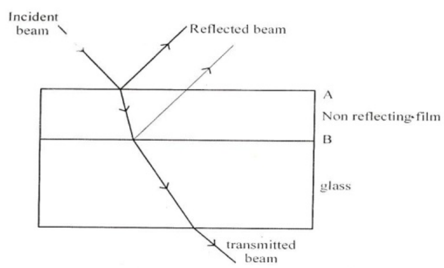
$$v = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$$

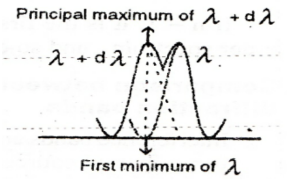
- **Fundamental frequency is directly proportional to the square root of Tension**
- **Fundamental frequency is inversely proportional to the length of vibrating string**
- **Fundamental frequency is inversely proportional to the square root** of linear **density**

Q3 **Explain Anti reflection coating**.

ans **This is an important application of thin film interference and it is used to reduce the loss of intensity of the incident beam of light by reflection . more and more intensity is lost if the number of reflection increases.**

The thickness of film t= $\lambda/4\mu$



		The loss of intensity due to reflection can be reduced by coating the reflecting surface with a suitable transparent dielectric material such as magnesium flouride. The refractive index of such material must be in between that of air and glass. Such a film is called non reflecting film.
	Q4	**Give Rayleigh's criteria for spectral resolution. Illustrate it with figure.**
	ans	**According to Rayleigh's criterion for resolution ,twoneighbouring spectral lines will be just resolved when the principal maximum of one in any order falls on the first minimum of the other in the same order**  ***Let*** **λandλ+dλ be the wavelengths corresponding to two neighbouring spectral lines of the same order. Then, the*two spectral lines are visible as separate* when the principal maximum ofwavelengtλ+dλ falls on the first minimum of wavelengthλof the same order.**
	Q5	**Write the physical significance of wave function**
	ans	**The quantity whose variation makes up the matter waves is called the wave functionΨ.The quantity that undergoes periodic changes of a body is Ψ.** **$\Psi = Ae^{i(kx-\omega t)}$** **Ψ is a complex valued function and itself has no physical meaning. The square of the absolute magnitude/ $\Psi/^2$ or Ψ Ψ*dxdydz gives the probability of finding the particle in the volume element dxdydz. We can obtain the physical properties of the system if we know the wave function.** **Since $\psi\psi^* dxdydz$ is proportional to the probability of finding the particle with in the volume element, the integral $\int \psi^* \psi dxdydz$ must be finite if the particle is somewhere there. If $\int \psi^* \psi dxdydz$ is zero, the particle does'nt exist and if it is infinity, the particle is everywhere simultaneously.**
	Q6	**Why do nanomaterials exhibit properties different from those of their classical counterparts?**
	ans	**The important factors that cause significant change in the properties of nano materials are** **• Increased surface to volume ratio** **• Quantum confinement effects** **Quantum confinement is the restricted motion of randomly moving electron in specific energy levels when the dimensions of a material approach the de Broglie wavelength of electron. Thus when the radius of the sphere decreases, its surface area to volume ratio increases. Hence as the particle size decreases a greater number of atoms are found at the surface compared to those inside. Thus nano material have more surface area per given volume compared with larger particles. It makes material more chemically reactive as chemical reaction occurs at surfaces.**

	Q7	**State and explain Ampere's circuital law.**
	ans	**Oersted observed that a magnetic field is always produced around a conductor carrying current. Amperes circuital law explains the relationship between the current and the magnetic field created by it.** **It state that the line integral of the magnetic flux density (B) for a closed path is equal to μ_0 times the net currenti enclosed by the path** $\boxed{\oint_C \underline{B} \cdot d\underline{l} = \mu_0 I_{encl}}$
	Q8	**State and explain Poynting's theorem.**
	ans	**The magnitude and direction of flow of energy per unit area per unit time in an Electromagnetic wave travelling in free space is represented by a vector known as Poynting vector.** **S=E X H unit is Watt per square meter.** **Poynting vector represents the instantaneous power density associated with the EM field at a given point.** $S=\frac{P}{A}$
	Q9	**What are high temperature superconductors? Give two examples.**
	ans	**In 1977 a high transition temperature 23 K was achieved using metallic compound of niobium and Germanium .But in 1986 Bedddnorz and Muller discovered La – Ba-CuO system of ceramic super conductors with a Tc of 34 K. Such superconductors with high critical temperature are called High Temperature Superconductors. High Temperature Super conductors are ceramic super conductors with a high transition temperature greater than 40 K.**
	Q10	**What are fibre optic sensors? Name two different types.**
	ans	**Optic fiber sensors are very sensitive devices used to measure physical quantities like temperature ,pressure, displacement ..etc very accurately with the help of optic fibers. Sensor has 3 components:** **1. A source of light** **2. A fiber coil** **3. Detector** **This is based on the based on the principle that the modulation of light takes place whenlight is transmitted through the fiber when subjected to the physical quantity.** **INTENSITY MODULATED SENSORS** **PHASE MODULATED SENSORS**
		PART B
		Answer any one full question from each module. Each question carries 14 marks
		Module 1
	Q11	**Obtain the differential equation of a damped harmonic oscillator and deduce the solution for over damped condition. Show the graphical variations of**

	a	**displacement with respect to time.**
	ans	**DAMPED HARMONIC OSCILLATION** **A harmonic oscillator in which the motion is damped by the action of an additional force is called a damped harmonic oscillator. Damped oscillations are oscillations under the action of resistive forces.** **Damping force is α to its velocity.** **ie Damping force = b** $\frac{dx}{dt}$ **'b' is called the damping constant. Including the damping force, the differential eq. for a damped harmonic oscillator is** $m\frac{d^2x}{dt^2} = -kx - b\frac{dx}{dt}$ $m\frac{d^2x}{dt^2} + kx + b\frac{dx}{dt} = 0$ $\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0 --------- (1)$ $Let\ \frac{b}{m} = 2\gamma \quad where\ \gamma\ is\ the\ damping\ coefft.$ $\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega^2 x = 0 --------- (2)$ *This is the differential eqn for damped harmonic oscillator* **Solutions to the Equation** **Assume the solution is of the form** $x = Ae^{\alpha t}$ $\frac{dx}{dt} = \alpha Ae^{\alpha t}$ $\frac{d^2x}{dt^2} = \alpha^2 Ae^{\alpha t} = \alpha^2 x$ *Putting these values in (2), we get* $\alpha^2 x + 2\gamma\alpha x + \omega^2 x = 0$ $ie\ \ \alpha^2 + 2\gamma\alpha + \omega^2 = 0$ $The\ solution \quad \alpha = \frac{-2\gamma \pm \sqrt{\left(4\gamma^2 - 4\omega^2\right)}}{2}$

$$x = Ae^{\left(-\gamma \pm \sqrt{(\gamma^2-\omega^2)}\right)t}$$

The solutions can be written as

$$x_1 = A_1 e^{\left(-\gamma + \sqrt{(\gamma^2-\omega^2)}\right)t} \quad and \quad x_2 = A_2 e^{\left(-\gamma - \sqrt{(\gamma^2-\omega^2)}\right)t}$$

$\therefore$ *The general solution is* $x = x_1 + x_2$

$$x = A_1 e^{\left(-\gamma + \sqrt{(\gamma^2-\omega^2)}\right)t} + A_2 e^{\left(-\gamma - \sqrt{(\gamma^2-\omega^2)}\right)t} \quad ----(3)$$

Where A_1 *and* A_2 *are constants which depends on the initial values of position and velocity.*

$'\gamma'$ *determines the behaviour of the system.*

<u>Case 1</u>: γ>ω

The roots of the auxiliary equation are real

$$\alpha_1 = -\gamma + \sqrt{(\gamma^2-\omega^2)}; \quad \alpha_2 = -\gamma - \sqrt{(\gamma^2-\omega^2)} ---(4)$$

The solution of equation (3) is

$$x = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} -----------(5)$$

Where A_1 and A_2 are constants.

Let the mass be given a displacement x_0 and then released ,So that

$$x = x_0, \quad \frac{dx}{dt} = 0 \ at \ t = 0$$

$$From\ eq(5)\ we\ get \quad x_0 = A_1 + A_2$$

$$and \quad A_1\alpha_1 + A_2\alpha_2 = 0$$

$$Simplifying, \ A_1 = -\frac{x_0\alpha_2}{\alpha_1 - \alpha_2} \quad and \quad A_2 = \frac{x_0\alpha_1}{\alpha_1 - \alpha_2} ---(6)$$

$$\therefore Eq(5) \ reduces \ to \quad x = \frac{x_0}{\alpha_1 - \alpha_2}\left(\alpha_1 e^{\alpha_2 t} - \alpha_2 e^{\alpha_1 t}\right)$$

This shows that x is always positive and decreases exponentially to zero without any oscillations.The motion is called over damped or dead beat motion

Displacement curve

a) overdamped

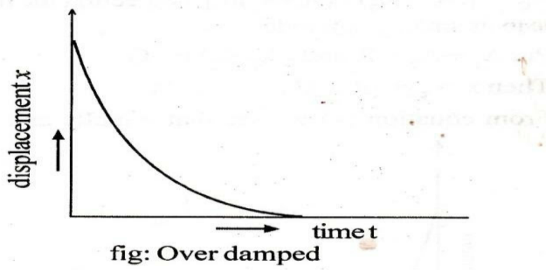
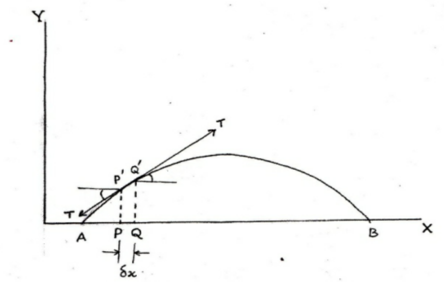


fig: Over damped

	b	**The equation of transverse wave travelling along a stretched string is given by,** **ψ (x,t)=10 sin (2πt-0.01πx) where ψ and x are in cm and t is in** **second. Find the amplitude, frequency, velocity and wave length**
	ans	(1) $\psi = 10 \sin(2\pi t - 0.01\pi x)$ $y = a \sin(kx - \omega t)$ $a = 10\ cm$, $k = 0.01\pi$ $\omega = 2\pi$ $v = \frac{\omega}{k} = \frac{2\pi}{0.01\pi} = 200\ cm/s$ $k = \frac{2\pi}{\lambda}$ $\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.01\pi} = 200\ cm$ $\omega = 2\pi\nu$ $\nu = \frac{\omega}{2\pi} = \frac{2\pi}{2\pi} = 1\ Hz$
	Q12 a	**Derive the differential equation for transverse wave in a stretched string and hence obtain the expression for velocity of the wave.**
	ans	**Consider an infinitely long thin and uniform string stretched between two points by a constant tension T along the X-axis. Let the string be slightly displaced aside along the Y-axis and released. Transverse vibrations are set up in the string.** **Consider a small element PQ of the string of lengthδx. The magnitude of tension in the string will be same everywhere since the string is perfectly flexible. The tension T acts tangentially at every point on the string. P'Q' is the displaced position of the string.**  **The tension at P' and Q' acts along the tangent making angles θ_1 andθ_2 with the horizontal. Resolving the tension along x- axis and y- axis,** **Thenet force on PQ acting X and Y direction are** $F_x = T\cos\theta_2 - T\cos\theta_1$ $F_y = T\sin\theta_2 - T\sin\theta_1$ **For small oscillations θ_1 and θ_2 are very small.** $\cos\theta_1 = 1$; $\cos\theta_2 = 1$ **Also** $\sin\theta_1 = \tan\theta_1$; $\sin\theta_2 = \tan\theta_2$ $(\tan\theta_1=\sin\theta_1 /\cos\theta_1 =\sin\theta_1)$ **Then** $F_x = 0$ $F_y = T\tan\theta_2 - T\tan\theta_1$

So the net force acting on the element δx in the displaced position is only along Y-axis.

$F_y = T(\tan\theta_2 - \tan\theta_1)$

If μ is the mass per unit length of the string (linear density) ,

mass of the element δx, m = μ δx

Acceleration = $\dfrac{\partial^2 y}{\partial t^2}$

According to the Newton's second law of motion

F=ma

F = μ δx $\dfrac{\partial^2 y}{\partial t^2}$

$$\mathrm{T}\left(\tan\theta_2 - \tan\theta_1\right) = \mu\,\delta x.\frac{\partial^2 y}{\partial t^2}$$

$$But \quad \tan\theta_1 = \left(\frac{\partial y}{\partial x}\right)_x \quad ; \quad \tan\theta_2 = \left(\frac{\partial y}{\partial x}\right)_{x+\partial x}$$

$$or \qquad \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu}\left[\frac{\tan\theta_2 - \tan\theta_1}{\delta x}\right]$$

$$ie \qquad \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu}\left[\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\partial x} - \left(\frac{\partial y}{\partial x}\right)_x}{\delta x}\right]$$

$$when \ \delta x --> 0$$

$$Lim\,\delta x --> 0 \left[\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\partial x} - \left(\frac{\partial y}{\partial x}\right)_x}{\delta x}\right] = \frac{\partial^2 y}{\partial x^2} --- (5)$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu}\frac{\partial^2 y}{\partial x^2} ---------- (6)$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T}\frac{\partial^2 y}{\partial t^2}$$

This is the differential equation of a vibrating string.

Comparing this equation with the standard wave equation

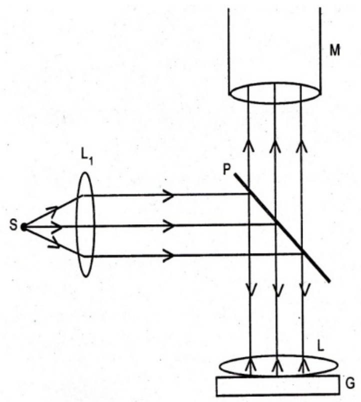
$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

$$\frac{1}{v^2} = \frac{m}{T}$$

$$v^2 = \frac{T}{m}$$

$$v = \sqrt{\frac{T}{m}}$$

This is the velocity of transverse wave on a stretched string.

		$v = \upsilon\lambda$ $\upsilon = \frac{v}{\lambda}$ $v = \frac{1}{\lambda}\sqrt{\frac{T}{m}}$ **This is the frequency of transverse waves developed on a stretched string.**
	b	**The frequency of a tuning fork is 200Hz. If the quality factor Q = 5 x 10^4 , find the time after which its amplitude becomes 1/2 of its initial value**.
	ans	$\nu = 200 Hz$ $Q = 5\times10^4$ $Q = \omega\tau$ $Q = 2\pi\nu\tau$ $\tau = \frac{Q}{2\pi\nu} = \frac{5\times10^4}{2\times\pi\times200} = 39.80 sec$ $A = A_0 e^{-\gamma t}$ $\frac{A_0}{2} = A_0 e^{-\gamma t}$ $\frac{1}{2} = e^{-\gamma t}$ $2 = e^{\gamma t}$ $\log_e 2 = \gamma t$ $t = \frac{\log_e 2}{\gamma} = \frac{0.69314}{0.0125} = 55.2 sec$ $\tau = \frac{1}{2\gamma}$ $\gamma = \frac{1}{2\tau} = \frac{1}{2\times39.8} = 0.0125$
Module 2		
	Q13 a	**Describe the experimental set up of Newton's ring arrangement. Derive an expression for wavelength of light used.**
	ans	**MEASUREMENT OF WAVELENGTH OF LIGHT**  **S is a sodium vapour lamp. Light from lamp is rendered parallel by a convex lens L_1 fall on the**

glass plate P kept inclined at 45 degree , gets reflected, and then falls normally on the convex lens L placed over the glass plate G. A pattern of bright and dark circular rings are observed through a microscope arranged vertically above the glass plate P. The microscope is focused so that the rings are clearly seen .

The cross wire of the microscope is kept at the central dark spot. Then by working the tangential screw of the microscope,thecross wire is moved towards the left so that the crosswire is tangential to the 20th dark ring on the left side. The main scale and vernier scale readings of the microscope are taken. Then by working the tangential screw , the cross wire is kept tangential to the18th 16th,14thetc dark rings up to the second dark ring on the left and the corresponding readings are taken. Then the cross wire is made tangential to the second dark ring on the right side. Readings are taken corresponding to the 2nd,4th etc….. 20th dark rings on the right side as before. The difference between the readings on the left and right of each ring gives the diameter D of the respective ring. Then D^2 is found out .Hence ($D_{n+k}^2 – D_n^2$) is calculated (k=10).

The radius of curvature R of the lower surface of the lens is found by Boy's method.For this ,the convex lens Lis placed in front of an illuminated wire gauze, with the marked surface away from the wire-gauze. With a black paper held behind the lens the position of the lens is adjusted so that a clear image of the wire-gauze is formed side by side with it. The distance 'd' between the lens and the wire-gauze is measured. This is repeated 3 times and the mean value of'd' is found out.

Then the focal length '*f*' of the convex lens L is determined by plane mirror method. For this,aplane mirror is held behind the lens and the position of the lens is adjusted so that a clear image of the wire is formed side by side with .The distance between the wire gauze and the convex lens is the focal length(f).Repeat the measurement three times and the average value of '*f*' is found out .

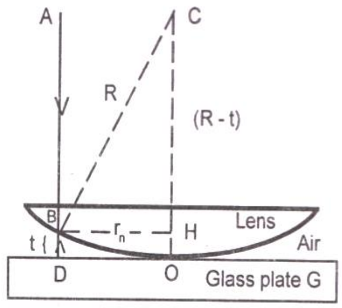
Then the radius of curvatureof the lens is found out using the formula

$$R = \frac{fd}{f-d}$$

The wave length of sodium light is hence calculated using the formula,

$$\lambda = \frac{D_{n+k}^2 - D_n^2}{4kR}$$

RADIUS OF THEnth DARK RING



Let C be the centre of curvature of the curved surface of the lens and R be its radius of curvature. Let nth dark ring be formed through B where the thickness of the air film is 't'. Let r_n be the radius of dark ring ,ieBH=r_n

The condition for the formation of the nth dark ring at D is,

$2\mu t \cos r = n\lambda$ ....(1)

where n = 0, 1, 2, 3 ....

Since the light is incident normally, r = o or Cos r = 1.

Also $\mu = 1$ for air.

$\therefore 2t = n\lambda$ ....(2)

From the right angled triangle CBH,

$CB^2 = CH^2 + BH^2$

ie, $R^2 = (R - t)^2 + r_n^2$

$= R^2 + t^2 - 2Rt + r_n^2$.

Since t is v ery small t^2 is negligible.

$\therefore R^2 = R^2 - 2Rt + r_n^2$

or $r_n^2 = 2Rt$ ....(3)

$2t = n\lambda$ from (2).

$\therefore r_n^2 = Rn\lambda$ ....(4)

or $r_n = \sqrt{Rn\lambda}$ ....(5)

Radius of the ring , r α $\sqrt{n}$

As n increases, the distance between the rings decreases. That is ,the rings come closer as we move away from the centre.

WAVELENGTH OF LIGHT

The radius of the n^{th} dark ring is

$$r_n = \sqrt{R n \lambda}$$

ie $$r_n^2 = R n \lambda$$

If D_n is the diameter of the n^{th} dark ring,

$$r_n = \frac{D_n}{2}$$

		or $r_n^2 = \frac{D_n^2}{4} = R\,n\,\lambda$ or $D_n^2 = 4Rn\lambda$ ....(9) If D_{n+k} is the diameter of the (n+k)th dark ring, $D_{n+k}^2 = 4R(n+k)\lambda$ ....(10) Then $D_{n+k}^2 - D_n^2 = 4Rk\lambda$. or $\lambda = \frac{D_{n+k}^2 - D_n^2}{4kR}$ ....(11) Thus, measuring D_{n+k}, D_n, and R, the wavelength λ can be calculated. Similarly, the radius of the nth bright ring, r_n, is given by $r_n^2 = R(2n-1)\frac{\lambda}{2}$ Then $r_n = \frac{D_n}{2}$ or $r_n^2 = \frac{D_n^2}{4} = R(2n-1)\frac{\lambda}{2}$ or $D_n^2 = 4R(2n-1)\frac{\lambda}{2}$ ....(12) $D_{n+k}^2 = 4R[2(n+k)-1]\frac{\lambda}{2}$ ....(13) $D_{n+k}^2 - D_n^2 = 4Rk\lambda$ $\lambda = \frac{D_{n+k}^2 - D_n^2}{4kR}$
	b	**Light of wavelength 6000A° falls normally on two glass plates enclosing a wedge shaped film. The plates touch at one end and are separated at 10cm from that end by a wire. If the bandwidth of the interference pattern is 0.05 mm, find diameter of the wire.**

ans

$l = 10\ cm$

$\beta = 0.05\ mm$

$$d = \frac{\lambda L}{2\beta} = \frac{6000\times10^{-10}\times 10\times10^{-2}}{2\times0.05\times10^{-3}} =$$

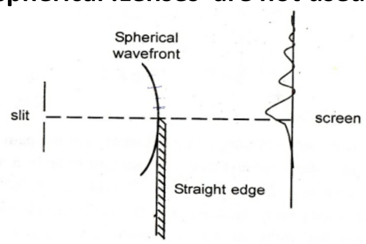
$$= 6\times10^{-4}\ m$$

Q14 a

Give any 3 differences between Fresnel and Fraunhofer classes of diffraction. Discuss diffraction due to grating and derive the grating equation for normal incidence.

ans

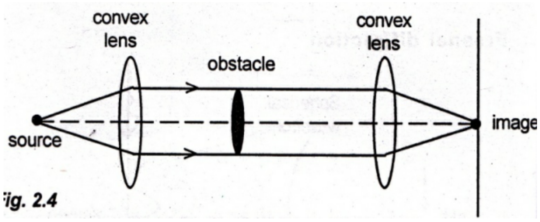
14. **) FRESNEL DIFFRACTION - Either the source of light or the screen or both are at finite distance from the obstacle causing diffraction. Wavefront falling on the obstacle is spherical .Lenses are not used.**



Example :Diffraction at a straight edge

FRAUNHOFER DIFFRACTION

The source of light and the screen are at infinite distance with respect to the obstacle causing diffraction. Wavefront falling on the obstacle is plane. Lenses are used

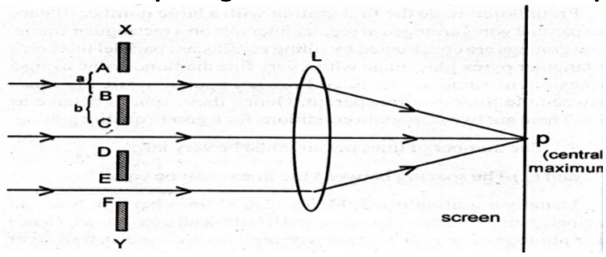


ig. 2.4

Example:Diffraction at a grating

PLANE TRASMISSION GRATING is a plane glass plate containing a large number of equidistant parallel lines drawn using a fine diamond point . The space between the lines acts as narrow slits through which light is transmitted. The lines are opaque to light. Grating is an arrangement of a large number of parallel slits of equal width separated by equal opaque spaces . Usually a grating has 5000 to 12000 lines per cm. There are two important conditions for a good quality grating.

1. **The number of lines per cm must be very large.**
2. **The spacing between the lines must be equal.**



Consider a plane transmission grating placed perpendicular to the plane

of paper. AB represents a slit and BC represents a line. Let 'a' be the width of each slit and 'b' be the width of each line. The distant (a+b) is called grating element or grating constant.

A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit. Each point of the wave front sends outsecondary waves in all direction. The straight and parallel waves from each point can be focussed on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called central maximum.

The position of central maximum is same for all the wavelength.The central maxima will have the same colour as the incident light.

Consider two waves diffracted from two points A and C of slit. They travels along Am and CN. Draw CK perpendicular to AM. Then the path difference between the two waves is AK.

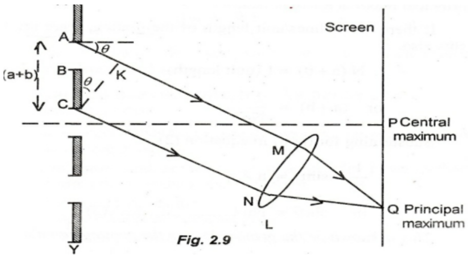


Fig. 2.9

From triangle ACK

$\sin\theta = AK/AC$

$AK = AC \sin\theta$

$AK = (a+b)\sin\theta$

If $(a+b)\sin\theta = n\lambda$ ....(1)

where $n=0,1,2,3.....$ two waves interfere constructively. This is called principle maximum. For different values of n , there are different values of θ.

If n=1, it is the first order principle maximum, if n=2, it is the second order principal maximum and so on. Thus on either side of central maximum ,a number of principal maxima are obtained.

If there are N lines/unit length of the grating , there are N slit also.

$N(a+b) = 1$ (unit length)

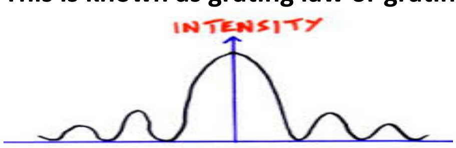
$(a+b) = 1/N$

Substitute in equation (1)

$1/N \sin\theta = n\lambda$

$\sin\theta = Nn\lambda$

This is known as grating law or grating equation.



b What is the highest order spectrum which may be seen with light of wavelength 5×10^{-5} cm by means of grating with 3000lines/cm?

ans

$\sin\theta = Nn\lambda$

$1 = Nn\lambda$

$n = \frac{1}{N\lambda} = \frac{1}{3000 \times 5 \times 10^{-5}} = 6.66$

$= 6$

Module 3

ans

Q15 a **Write the differential equation for a particle in a one dimensional box and obtain the possible energy values and normalized wave functions**

ans

ONE DIMENSIONAL INFINITE SQUARE WELL POTENTIAL

Consider the motion of a particle of mass m confined to move in a one dimensional square well of infinite depth.

Since there is no interaction between the particle and wall, the potential energy of the particle is taken to be zero. It is very clear that potential energy does not depend on time and we are considering only time independent Schrodinger equation for the solution. The problem is one dimensional and the Schrodinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\,\psi = 0$$

since $V = 0$, the equation becomes

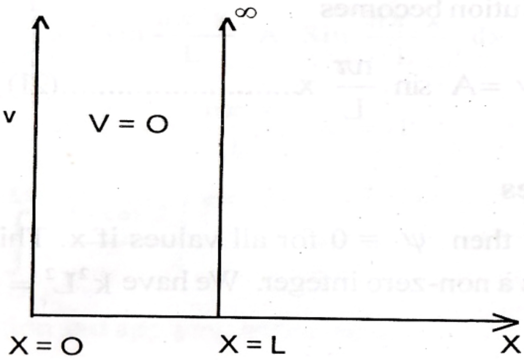
$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\,\psi = 0 \quad \text{or}$$

$$\frac{d^2\psi}{dx^2} + k^2\,\psi = 0 \quad \text{where } k^2 = \frac{2mE}{\hbar^2} \ldots\ldots\ldots\ldots\ldots\ldots(17)$$

The solution of the above differential equation is of the form

$$\psi = A\sin kx + B\cos kx \ldots\ldots\ldots\ldots(18)$$

The solution has to be well behaved wave function. Since the particle is inside a square well of infinite depth, it is impossible to find the particle outside. ie ψ must be zero for all points outside the square well.



$\psi = 0$ for $x < 0$ and

$\psi = 0$ for $x > L$

This is possible only if $\psi = 0$ at $x = 0$ and $x = L$ as demanded by continuity condition.

Applying first condition on equation 18, we get

$$0 = A \sin 0 + B \cos 0$$

ie B = O

So solution reduces to

$$\psi = A \sin kx \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (19)$$

Using the next condition $\psi = 0$ at $x = L$, we get

$$O = A \sin kL$$

There are two possibilities. Either $A = 0$ or $\sin kL = o$

A cannot be zero, since the wavefunction cannot exist. So we have the other possibility

Sin kL = o

ie $kL = n\pi \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (20)$

where n is an integer.

So the solution becomes

$$\psi = A \sin \frac{n\pi}{L} x \ldots\ldots\ldots\ldots\ldots\ldots\ldots (21)$$

Eigen Values

If $n = 0$, then $\psi = 0$ for all values if x. This is ruled out. Therefore n is a non-zero integer. We have $k^2L^2 = n^2\pi^2$ on using equation (17)

$k^2 = 2mE/\hbar^2$

$$\frac{2mE}{\hbar^2}L^2 = n^2\pi^2$$

$$E = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad \text{or}$$

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \;.\; \text{ie} \;\; E_n = \frac{n^2h^2}{8mL^2}$$

E_n as the energy corresponding to n.The quantity n is called quantum number. The different values of energy for n are called Eigen values. Since n is restricted, the particle cannot have any value of energy but will be restricted to certain values.

Eigen wave functions

The wave functions associated with different Eigen values of energy are called Eigen Functions.
By applying the normalization condition,

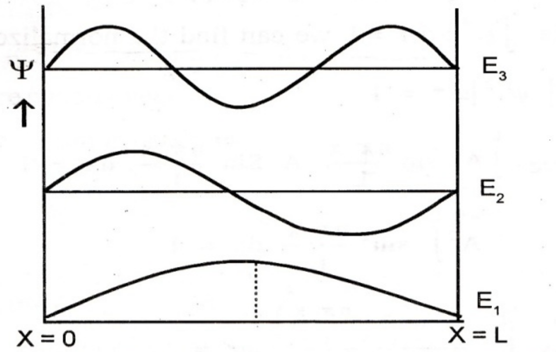
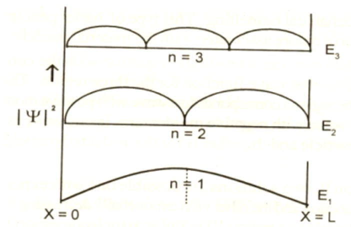
$$\int \psi^* \psi dxdydz = 1$$

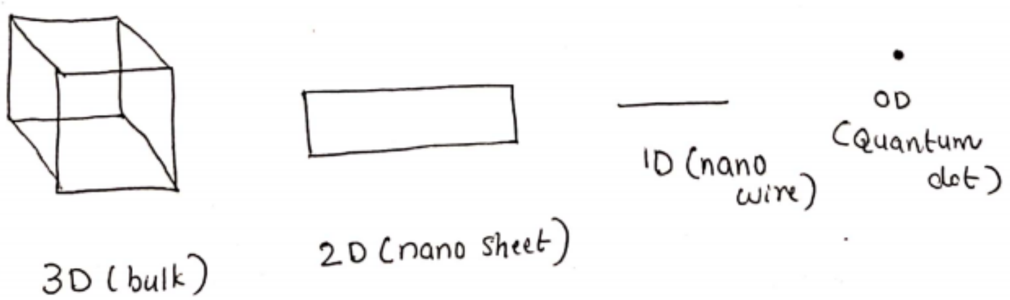
$$\text{So we have} \int_0^L A \sin\frac{n\pi x}{L} \; A \sin\frac{n\pi x}{L} \, dx = 1$$

$$A^2 \int_0^L \sin^2\frac{n\pi x}{L} \, dx = 1$$

$$A^2 \int_0^L \frac{1 - \cos 2\left(\frac{n\pi x}{L}\right)}{2} \, dx = 1.$$

On integration and applying limits we get

		$A^2 \frac{L}{2} = 1 \qquad \text{or } A = \sqrt{\frac{2}{L}}$ So the normalised wave function is $\psi_n(x) = \sqrt{\frac{2}{L}} \text{Sin}\left(\frac{n\pi x}{L}\right)$ where n = 1, 2, 3..........(22) For the ground state n = 1, the wave function is given by $\psi_1 = \sqrt{\frac{2}{L}} \text{ Sin } \frac{\pi x}{L}$ Similarly the wave function for the first two excited states are given by $\psi_2 = \sqrt{\frac{2}{L}} \text{ Sin } \frac{2\pi x}{L} \qquad \psi_3 = \sqrt{\frac{2}{L}} \text{ Sin } \frac{3\pi x}{L}$ These wave functions associated with different energy are called Eigen wave functions.  *Fig. 2 wave function of first three energy levels* 
	b	**Calculate the quantum number associated with a marble of mass 10 gm trapped to move with speed 1m/s in a one dimensional box of width 20 cm**

	ans	$E = \frac{n^2 h^2}{8mL^2} =$ $n^2 = \frac{E\,8\,mL^2}{h^2}$ $= \frac{8 \times 10 \times 10^{-3} \times (20 \times 10^{-2})^2 \times 5 \times 10^{-3}}{(6.626 \times 10^{-34})^2}$ $= \underline{6 \times 10^{30}}$ $E = \frac{1}{2} m v^2$ $= \frac{1}{2} \times 10 \times 10^{-3} \, 1^2$ $= \underline{5 \times 10^{-3}}$
	Q16 a	**Explain the following** **(i) Nanomaterials (ii) Nano sheets (iii) Nano wires and (iv) Quantum dots**
	ans	**Nanostructured materials** **The material which have atleast one dimension in the order of nanometer are called nanostructured materials** **Based on number of dimensions that are confined, nano structure is classified in to 4 categories** **1. Zero dimensional (0D)nano structured material:- If all the three dimensions become nanometer range, we obtain a Zero dimensional nano structure called nano particles** **2. One dimensional (1D)nano structured material:- If two dimensions are restricted and one dimension remains large, then the resulting structure is one dimensional nano structure called nanowires,nano rode and nano tube** **3. Two dimensional (2D)nano structured material:-If one dimension is reduced to the nano range while the other two dimensions remain large, then a two dimensional nano structured material are obtained (nano sheet)** **4. Three dimensional(3D) nanomaterial:-all the 3 dimensions become large(bundles of nano wires)** • **Zero dimensional (0D)nano structured material - 3 D confinement** • **One dimensional (1D)nano structured material - 2 D confinement** • **Two dimensional (2D)nano structured material - 1 D confinement** • **Three dimensional(3D) nano material – 0 D confinement**  **Quantum sheets**

		Quantum sheets are two dimensional structure in which quantum confinement act only in one direction. Quantum confinement act along the thickness of the film. Electrons are free to move in xy plane of the film. So quantum confinement is in one dimension and are free to move in 2 dimensions. They are now used to make semiconductor lasers,LEDs and solar cell . **Quantum wires** **A nano wire or quantum wire is a wire of dimensions of the order of nanometers. Quantum wire is a one dimensional structure. Quantum wires are formed when 2 dimensions of the system are confined ie by making a thin wire of semiconductor. Two direction have quantum confinement. It is only free to move along one dimension that is along the wire. A quantum wire is an electrically conducting wire in which quantum effects influence the transport properties. Such effects appear in the dimension of Nano meters. So they are called Nano wires. Electrical conductance of wire is found to be quantized in multiples of $2e^2/h$. quantum wires can be used as an electron wave guide. It is used to make high speed lasers.** **Quantum dots** **Quantum dots are zero dimensional structure in which the electron is confined in all three dimensions. Their energy states are quantized in all three directions. consider a tiny dot of semi conductor material then we have a 3D quantum confinement of charge carriers. That is there are no freely moving electrons or holes in any direction .This is called "quantum dot". Although this quantum dot consists of thousands of atoms it behaves like a single atom with discrete energy levels rather than many atoms. When the size of the crystal become so small that it approaches the size of exitonbohr radius, then the electron energy levels will change from continuous to discrete values. That is their is a finite separation between energy levels. The energy gap increases with the decreases in the diameter of the dot. So quantum dot can emit any colour from blue to red depending on the size of the dot.**
	b	**What are the conditions to be satisfied by a well behaved wavefunction? Write its normalization condition.**
	ans	**REQUIREMENTS OF WAVE FUNCTION** 1. **Ψ must be finite, single valued and continuous.** 2. **$\partial \Psi/\partial x$ $\partial \Psi/\partial y$, $\partial \Psi/\partial z$ must also be continuous and single valued everywhere.** 3. **Ψ can be normalized** **Since the probability of finding the particle in the volume element is a surety, then** $\int \psi^* \psi dxdydz = 1$ **The wave function which obeys this condition is called normalized wave function.**
Module 4		
	Q17 a	**Distinguish between paramagnetic and diamagnetic substances with two examples for each**
	ans	**DIAMAGNETISM** • **Diamagnetic materials are substances which have tendency to move from stronger to weaker magnetic field when placed in non uniform magnetic field.** • **Repelled by the applied magnetic field.**

- If a rod of diamagnetic material is suspended in uniform field it will orient perpendicular to the field
- Material having even atomic number
- Atomic shells are filled and have no unpaired electron
- Magnetic moments are equal and opposite for a pair of electrons
- The atoms or molecules possess zero magnetic moment.

    When magnetic field is applied,

    - Electrons having orbital moment in the same direction of applied Magnetic Field slow downand those in the opposite direction speed up.
    - The orbital motion gets distorted.
    - Net magnetic moment is produced which opposes the applied field.
    - Result in repulsion
    - Posses' very weak susceptibility and is negative.
    - X is independent of temperature.
    - Permeability is less than one
    - Examples are Copper, Bismuth, Lead,water.

Langevin theory

Langevin theoretically calculated the susceptibility of a diamagnetic material, taking the magnetic moment produced due to orbital motion of electron.

Susceptibility, $\chi = -\frac{\mu e^2 n z}{6m}\langle r^2\rangle$

μ_0-permeability of free space

n - No of atoms per unit volume

z - Atomic number

e - Electronic charge

m - Mass of electrons

$\langle r^2\rangle$ - Mean square distance of electron from the nucleus

PARAMAGNETISM

- Paramagnetic materials are weakly magnetised in an externally applied magnetic field.
- Paramagnetic material show tendency to move from weak to strong magnetic field when placed in a non uniform magnetic field.
- Paramagnetism is due to the presence of unpaired electrons in atoms.
- Electrons undergoes spin motion and orbital motion and produces spin magnetic moment and orbital magnetic moment.
- These moments are randomly oriented and exhibit no magnetic moment

    When magnetic field is applied,
- The electrons spins tends to align with the applied field resulting in magnetic moment
- Induced field will be in the direction of applied field;but weak
- Free electron magnetism is weak and is called pauliPara magnetism
- They do not retain any magnetism in the absence of an external magnetic field.
- Susceptibility is low andpositive.
- Paramagnetic susceptibility depends on temperature
- Examples: O_2,Ne,

CURIE'S LAW

		According to curies law, the susceptibility of a paramagnetic material is inversely proportional to its temperature. $\chi \propto 1$ $\chi = \frac{C}{T}$ **C is called curie's constant and is given by** $C = \frac{\mu_0 n \mu^2}{3k}$ **μ_0= Permeability of freespace. n = no.of atoms per unit volume.** **μ= total magnetic moment due to spin and orbital motion, k=Boltzmann constant.**
	b	**Calculate induced emf and current in a closed circuit at time, t = 3s, if the magnetic flux through it varies with time obeying the equation $\phi = t^3+2t^2+5t$. The resistance in the circuit is 4 Ω.**
	ans	$\mathcal{E}$, I ? $t = 3$ sec, $R = 4\,\Omega$ $\phi = t^3+2t^2+5t$ $\mathcal{E} = -\frac{d\phi}{dt} = -\frac{d}{dt}(t^3+2t^2+5t)$ $= 3t^2+4t+5$ $= 3\times(3)^2+4\times3+5$ $= 27+12+5$ $= \underline{44\,V}$ $V = IR$ $I = \frac{V}{R}$ $= \frac{44}{4}$ $= \underline{11\,A}$
	Q18 a	**Starting from basic laws of electricity and magnetism, derive Maxwell's equations**.
	ans	**MAXWELL'S FIRST EQUATIONS** **Maxwell's First equation is the statement of Gauss's Law in electrostatics, it is also called Electric flux equation** **According to Gauss's law in electrostatics, The total normal electric flux through a closed surface is equal to $\frac{1}{\varepsilon_0}$ times the total charge enclosed by the surface.** **That is $\phi_E = \frac{q}{\varepsilon_0}$** $\int_s E.ds = \frac{q}{\varepsilon_0}$

$$q = \int_v \rho dv$$

$$\int_s E.ds = \frac{1}{\varepsilon_0}\int_v \rho dv$$

$$\int \varepsilon_0 E.ds = \int \rho dv$$

D=$\epsilon_0 E$ is called electric displacement vector

$$\int_s D.ds = \int_v \rho dv$$

Applying Gauss theorm

$$\int_v \nabla.Ddv = \int_v \rho dv$$

To hold this equation the integrands must be equal.

$\nabla.D = \rho$**…………………………Maxwell's First equation**

MAXWELL'S SECOND EQUATIONS

Second equation explains Gauss Law in Magnetism

By Gauss's law in magnetism the net magnetic flux emerging through any closed surface is zero.

Then$\int_s B.ds = 0$

By using Gauss theorm,

$$\int_v \nabla.\mathrm{B}\ \mathrm{dv} = 0$$

$\nabla.\mathrm{B} = 0$**……… Maxwell's second equation**

For a magnetic dipole, in any closed surface the magnetic flux directed inwards towards South Pole will be equal to the flux outward from North Pole. The net flux will be zero for a dipole.

MAXWELL'S THIRD EQUATIONS

Third equation describes Faraday's law of electromagnetic Induction

According to Faradays law of electromagnetic Induction,

Induced emf e=$-\frac{\partial\phi}{\partial t}$

ϕ is magnetic flux

e=$-\frac{\partial}{\partial t}\int B.ds$ **…………….(1)**

We know that e=$\int E.dl$ **…………(2)**

$\int_l E.dl = -\frac{\partial}{\partial t}\int_s B.ds$

Using Stokes theorm

$\int_s \nabla X E.ds = -\int_s \frac{\partial}{\partial t}B.ds$

To hold this equation Integrands must be equal

$\nabla X E = -\frac{\partial B}{\partial t}$ **………………………Maxwell's third equation**

MAXWELL'S FOURTH EQUATIONS

Fourth equation is the statement of modified Ampere's circuital law and is the equation for induced magnetic field

According to amperes circuital law the magnetic induction around a closed current carrying loop is equal to μ_0 times the current I flowing in the loop.

Thus $\int B.dl = \mu_0 I$

$\int \mu_0 H.dl = \mu_0 I$

$\int H.dl = I$……………….(1)

If J is the current density of the loop,

then current

I=$\int J.ds$ ……………….(2)

From eqn 1 and 2

$$\int_l H.dl = \int_s J.ds$$

Using Stokes theorem

$$\int_s \nabla X H.ds = \int_s J.ds$$

$\nabla X H$ **=J ……………………(4)**

From eqn of continuity

$\nabla.J = -\frac{\partial \rho}{\partial t}$ **…………………………(5)**

Take divergent of eqn (4)

$\nabla.\nabla X H = \nabla.J$

But $\nabla.\nabla X H = 0$ Since divergent of curl of any vector will be zero.

Hence $\nabla.J=0$ ………………(6)

Here there is incompatibility in eqn 5 and 6

Hence Maxwell introduced a new term J_D called displacement current density

$\nabla X H = J + J_D$ **…………………(7)**

Now take divergent on both sides of eqn (7)

$\nabla.\nabla X H = \nabla.J + \nabla.J_D$

$0 = \nabla.J + \nabla.J_D$

$\nabla.J = -\nabla.J_D$

From eqn 5

$-\frac{\partial \rho}{\partial t} = -\nabla.J_D$

From Maxwell's first law

$\rho = \nabla.D$

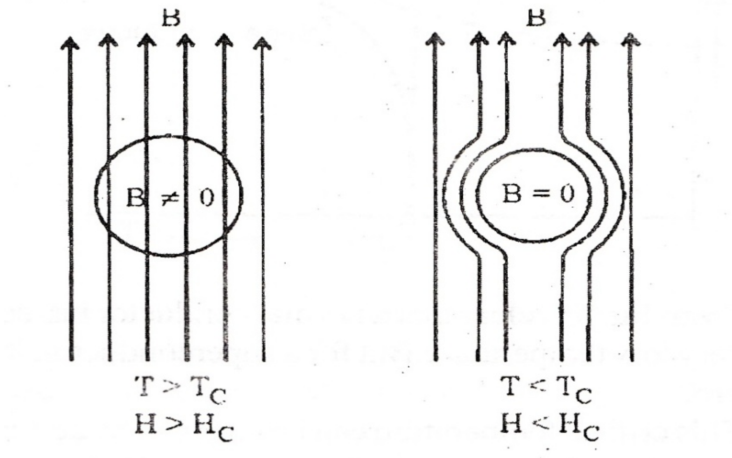
$\frac{\partial}{\partial t}\nabla.D = \nabla.J_D$

$\nabla.\frac{\partial D}{\partial t} = \nabla.J_D$

$\frac{\partial D}{\partial t} = J_D$

now substitute this value in eqn7

$\nabla X H = J + \frac{\partial D}{\partial t}$ **……………….This is Maxwell's fourth equation**

	b	**If $\phi(x, y, z) = 4x^2y - y^3z^2$, find the gradient of ϕ at the point (1,-1,-1).**
	ans	$\nabla\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$ $= \hat{i}\left[8xy\right] + \hat{j}\left[4x^2 - 3y^2z^2\right] + \hat{k}\left[-2y^3z\right]$ $\left[\nabla\phi\right]_{at\ (1,-1,-1)}$ $= \hat{i}\left[-8\right] + \hat{j}\left[4-3\right] + \hat{k}\left[-2\cdot -1\cdot -1\right]$ $= \underline{\underline{-8\hat{i} + \hat{j} - 2\hat{k}}}$
		Module 5
	Q19 a	**Explain Meissner effect in superconductivity. Distinguish between Type I and Type II superconductors with appropriate diagrams and examples**
	ans	**All the magnetic lines have been expelled from the interior of the specimen when it is cooled below the transition temperature in the magnetic field.** **The property of expulsion of magnetic flux density from the interior of a superconducting material during the transition from normal state to superconducting state is called Meissner effect.**  **When a specimen is placed in a magnetic field H, magnetic lines are passing through it.** $B = \mu_0(H+M)$ **Where,** **B = flux density inside the specimen** **μ_0 = Permeability** **M = magnetization** **When the specimen is cooled below the transition temperature, all magnetic lines are**

cancelled from the specimen.

Since B=0 , $\mu_0(H+M)=0$

H = -M

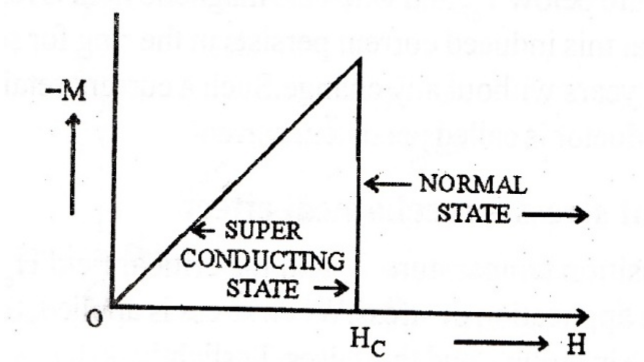
Magnetic susceptibility χ = M/H = -1

Hence the superconductor becomes a perfect diamagnet at the transition temperature
Superconductors are divided into 2 types depending on the magnetic properties. They are

1. **Type I or Soft superconductor**
2. **Type II or Hard superconductor**

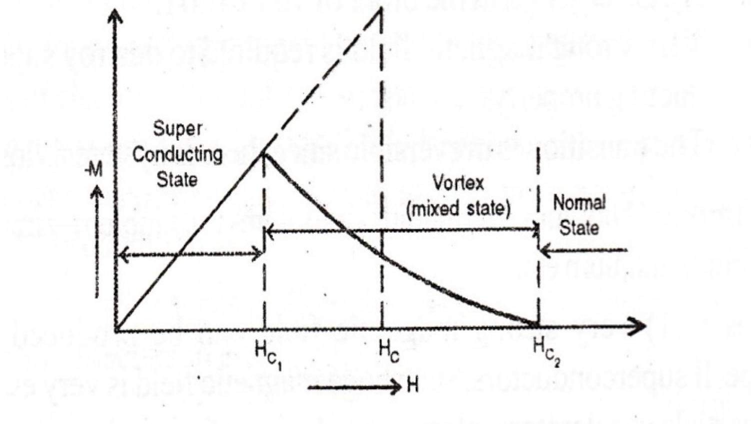
Type I or soft superconductors

When a superconductor is placed in a magnetic field H,the intensity of magnetization M is induced in it. When H increases, M also increases. The variation of M with H is shown in fig.

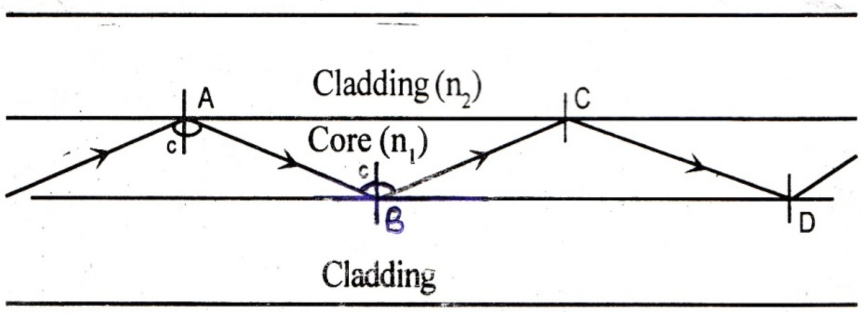


Up to the critical field Hc, magnetization increases proportional to the applied field and it abruptly drops to zero at the critical field. Above Hc it behaves as normal conductor. Below Hc, it is in the superconducting state. Up to Hc, it behaves like a diamagnet and strictly obeys Meissner effect.
Type of superconductors for which M becomes zero abruptly when H=Hc are called Type 1superconductors. The transition from superconducting state to normal state is very abrupt. The critical field Hc is very small in the order of 0 .1 or 0.2 Tesla. Therefore it is easy to change type I superconductor in to a normal conductor. Hence they are also known as soft superconductors. The transition is reversible.
Eg: Al, Pb,Indium and almost all pure metals.

Type 2 superconductors



		In this type there are two critical field,Lower critical field $Hc_1$and upper critical fields Hc_2. Up to Hc_1 magnetism is proportional to the applied field and it behaves like Type 1 superconductors. It strictly obeys Meissner effect up to Hc_1. In between $Hc_1$and Hc_2 magnetization decreases gradually.At Hc_2, the magnetisation vanishes completely and external field penetrates, completely destroying the superconductivity .Above Hc_2material behaves as normal conductor.Between Hc_1 and Hc_2material is in a mixed state (it behaves partially conducting as well as partially superconducting) known as vortex state. It does not strictly obey meissner effect in this region. **Transition from superconducting state to normal state is very slow and gradual.The fieldHc_2 value is very high in the order of 10T to 20T. A strong magnetic field is required to convert it into a normal conductor. Theyare known as hard superconductors. The transition is irreversible. Very strong magnetic field can be produced using type II superconductors.** **Eg: Niobium, Germanium and all alloys**
	b	**Give any four applications of superconductivity.**
	ans	**It has a wide range of applications** **1. Superconductors are used to produce a very strong and powerful magnetic field in the order of 20T at a low cost.** **a) To bend and guide the charged particles in particle accelerators , cyclotrons etc.** **b) Used for controlling and focusing high temperature plasma for the controlled nuclear fusion .** **c) Used in magnetic levitation –Maglev.** **d) Superconducting magnets are used to produce very efficient ore separating machines.** **2) Medical field** **a) MRI: The most important medical application of superconductivity is MRI.MRI is freefrom harmful effects of other radiations like X-rays.** **b) Superconducting magnetic field is used to remove tumourcells from the healthy cells** **c) A groupof squids is used for the diagnosis of epilepsy** **d) Using superconducting susceptometer the iron content in the body can be estimated .** **e) Squids are used to measure minute magnetic fields produced from heart and brain very accurately and used for pathological analysis of their functions.**
	Q20 a	**Explain the propagation of light through an optical fibre. Distinguish between step index and graded index fibres.**
	ans	**PROPAGATION OF LIGHT WAVES THROUGH AN OPTIC FIBRE**



When a ray of light is incident from core tothe cladding at an angle of incidence i greater than the critical angle at A,it undergoes total internal reflection at the core - cladding interface. So the ray rebounds along AB. It is again incident at B at an angle greater than the critical angle on the core - cladding interface. So it again undergoes total internal reflection and the ray is travelling along BC. The ray is undergoing successive total internal reflections at points on the core- cladding interface. Thus the wave is propagated through the optical fibre in a zig-zag manner by the multiple total internal reflection principle without any loss of energy.

CLASSIFICATION OF OPTIC FIBRE

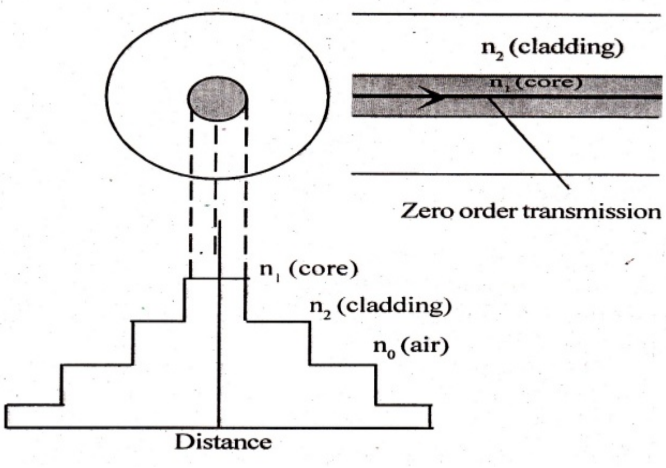
Optic fibre are generally classified into two groups depending on the refractive indices of core . They are

1. **Step index fibre**
2. **Graded index fibre**

STEP INDEX SINGLE MODE FIBER

The refractive index of core and cladding(n_1&n_2)are constants. $n_1 > n_2$.There is a sudden decrease of refractive index at the core- cladding interface.

INDEX PROFILE: A graph drawn with refractive index on y axis & the distance from the axis of the core on x axis.

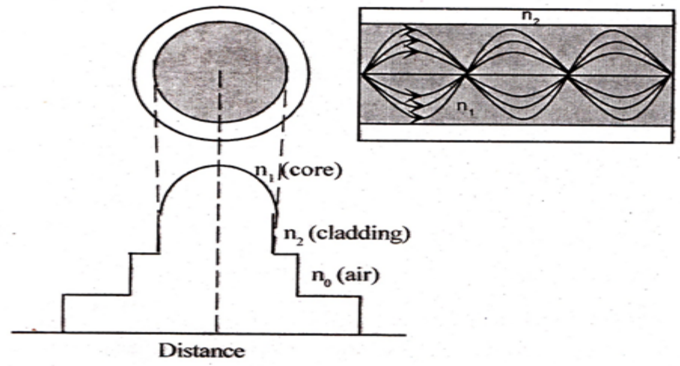


The refractive index profile is in the shape of a step and hence it is called step index fibre. This is reflective type fibre. Since the core is thin it supports only one mode for

propagation. So this is called single mode fibre. The light is propagated almost along the axis of the core. This is called zero order mode of transmission.

GRADED INDEX MULTI MODE FIBER

Refractive index of core (n_1) is not constant. But n_2 is a constant. $n_1 > n_2$ .n_1 is varying. It is maximum along the axis of the core, it decreases radially outwards with the distance from the axis and it is minimum at the core boundary. Ie refractive index is graded. Index profile is in the shape of a parabolic curve



Graded index fibre has a very thick core. Since the core is thick, it permits a large number of modes and hence it is a multi modefibre. This is refractive type fibre.

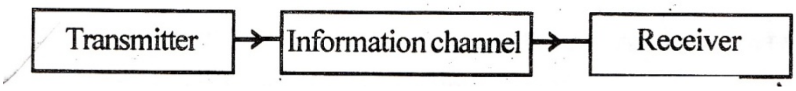
STEP INDEX SINGLE MODE FIBER	GRADED INDEX MULTI MODE FIBER
Refractive index of core and cladding(n_1&n_2)are constants.	Refractive index of core (n1) is not constant. But n2 is a constant
The refractive index profile is in the shape of a step	Index profile is in the shape of a parabolic curve
The core is thin. The light is propagated almost along the axis of the core	The core is thick. A large no. of rays are travelling along smooth parabolic curves.
It permit only one mode	Maximum number of modes that can be propagated is $V^2/4$
There is no pulse broadening effect and no intermodal dispersion.	No pulse broadening effect
This is reflective type	This is refractive type
The rays are travelling in a zig-zag manner	Rays are travelling along smooth paraboliccurves
Less expensive	Highly expensive
Band width is high andused for large distance communication	Band width is high andused for large distance communication

b **Explain the propagation of light through an optical fibre. Distinguish between step index and graded index fibres.**

ans

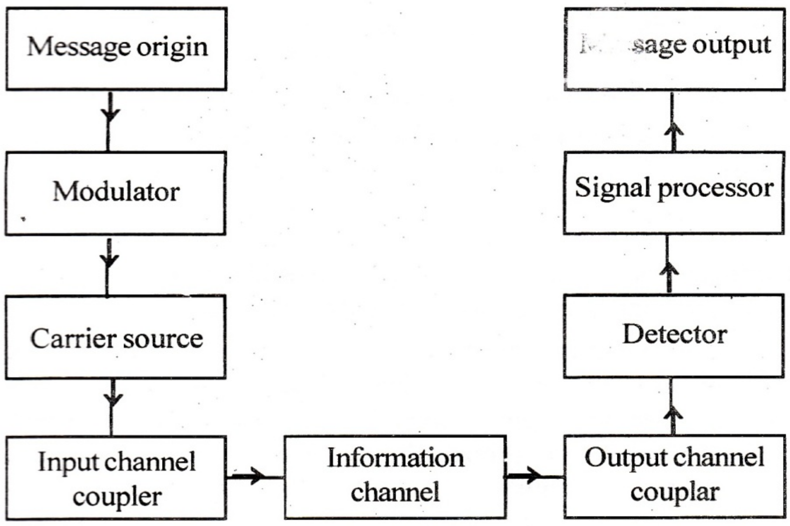
One of the important application of optic fiber is in the field of communication system. It is basically consists of 3 sections

1. **Transmitter: Converts electrical signal into optical signal**
2. **Information channel: Provides a path or passage to transmit the optical signals from transmitter to receiver**
3. **Receiver: Receives optical signals and converts back into electrical signals**



BLOCK DIAGRAM AND FUNCTION OF EACH COMPONENT

The message origin, modulator, carrier source, input channel coupler together constitute transmitter. Optic fibres are used as the information channel. Output channel coupler, detector, signal processor and message output constitute receiver.



1. MESSAGE ORIGIN

Converts all non electrical messages to electrical signals using transducer. Eg; microphone converts sound energy to electrical energy

2. MODULATOR

Imposing a message on a carrier wave for propagations is called modulation

At first it converts electrical message into proper format.

Secondly it imposes this format on a carrier wave for propagation.

There are 2 types of modulation. Analog and digital modulation. In analogue modulation message is transmitted in a continuous manner and in digital modulation message is transmitted in discrete manner with the help of binary digits.

Digital modulation is preferred for long distance communication.

3. CARRIER SOURCE

Carrier source produces carrier waves on which the messages are transmitted. In fiber communication system light waves are the carrier waves.LED or Laser diodes are used to generate stable and monochromatic waves . The information is imposed on light waves.

4. INPUT CHANNEL COUPLER

This directs modulated light waves into the information channels.

In the case of radio or television broadcasting systems antenna delivers radio frequency waves into atmosphere for propagation. Here antenna behaves as input channel coupler

5. INFORMATION CHANNEL

This is a path to transmit the information from transmitter to receiver .Here very fine and long optic fibres are used as information channel. Modulated light signals are transmitted through optic fibre by principle of total internal reflection.

6. OUTPUT CHANNEL COUPLER

This directs the modulated light signals from the information channel to the detector .Eg. Antenna in radio and television broadcasting systems.

7. DETECTOR

This detects and separates the messages from the modulated signals. ie, demodulation takes place. Here light signals are converted into electric current using photo detector.

8. SIGNAL PROCESSOR

This filters and selects the required frequency from waves. The selected frequency is amplified. The unwanted frequency is filtered out.

9. MESSAGE OUTPUT

Here the original message is reproduced from the signals. The electrical pulses are converted into sound waves in the case of audio systems. Cathode ray tubes and proper transducers are used for this.